

ENGINEERING TRIPOS PART IA 2014

Paper 3 Electrical and Information Engineering

Solutions

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SECTION A

1 (long)

Thevenin – replace any arbitrary linear circuit by a voltage source and an internal impedance, where the voltage source equals the open circuit voltage, and the internal impedance equals the open circuit voltage divided by the short circuit current.

Loop analysis is to work out voltages and currents around a network using a series of circulating currents, obeying nodes.

(b) $Z_1/Z_2 = Z_3/Z_4$ at balance, so no current through the meter.

Put Z_3 on bottom line, to make formulae nicer.

$$Z_2/Z_1 = Z_4/Z_3$$

$$(j\omega L + R_2)/R_1 = R_4/Z_3$$

$$1/Z_3 = 1/R_3 + j\omega C \quad !$$

$$(j\omega L/R_1) + (R_2/R_1) = R_4 \cdot (1/R_3 + j\omega C) = (R_4/R_3) + j\omega C R_4$$

Equate real and imaginary parts separately,

$$R_2/R_1 = R_4/R_3, \quad \text{or} \quad R_1 \cdot R_4 = R_2 \cdot R_3$$

$$j\omega L/R_1 = j\omega C R_4$$

$$\omega \text{ cancels! } L/C = R_1 \cdot R_4$$

(c)

$$Z_2/Z_1 = Z_4/Z_3 \quad \text{ensure } Z_3 \text{ on bottom line}$$

$$R_2/R_1 = (R_4 + 1/j\omega C) \cdot (1/R_3 + j\omega C)$$

$$= (R_4/R_3) + 1 + (1/j\omega C R_3) + j\omega C R_4$$

so, equate real and imag parts separately,

$$R_2/R_1 = (R_4 + R_3)/R_3$$

$$1/j\omega C R_3 = -j\omega C R_4$$

$$1 = \omega^2 C^2 R_3 R_4$$

$$\omega = \frac{1}{C \sqrt{R_3 \cdot R_4}}$$

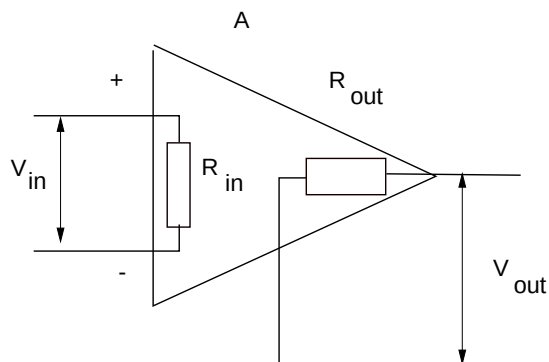
2 (long)

(a)

neg feedback has the advantages of

1. well defined gain, set by resistors.
2. extended frequency range
3. reduced distortion
4. independence from internal design of the op amp.
5. lowered output resistance (depending on design)
6. increased input resistance (depending on design)
7. opposite of these if you want current amplifier

(b)



(c)

The point here is not to give an involved formula for the gain, but an approximate numerical value.

$$\text{Gain} = (R_1 + R_2)/R_1 = 10.$$

$$A = 10^4, \beta = 1/10, \text{ so } A\beta = 10^3$$

$$R_{out} = 0 \text{ ohm.}$$

For impedance at V_2 ,

$$1/R = 1/75 + 1/150,000 \sim 1/75$$

$$R = 75 \text{ ohm.}$$

$$\text{Frequency response, } f = 1/2\pi CR_3 = 1/(2\pi \cdot 10^{-10} \cdot 75) = 10^8/(1.5\pi) \sim 20 \text{ MHz.}$$

3 (short)

$$4/20 = 10/(R1 + 10)$$

$$\text{so } R1 = 40 \text{ k}\Omega$$

$$I_{ds} = 4, \text{ V across } R2 = 10 \text{ V,}$$

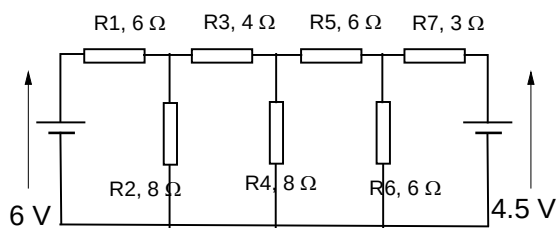
$$\text{so } R2 = 10/4 = 2.5 \text{ k}\Omega.$$

Gain needs R_{out}

$$1/R_{out} = 1/2.5 + 1/30 = 13/30. \quad R_{out} = 30/13 \text{ k}\Omega.$$

$$G = -5.30/13 = -150/13 = -11.53.$$

4 (short)



Want current thru 6V battery, so work from right.

Using Thevenin repeatedly, replace 4.5V and $R6 \parallel R7$ with R_a and V_a

$$1/R_a = 1/3 + 1/6 = 3/6, \text{ so } R_a = 2.$$

$$V_a = 6/(3 + 6) \cdot 4.5 = 3V.$$

$$R5' = 6 + 2 = 8$$

Replace $R5' \parallel R4$ and 3 V with R_b and V_b

$$1/R_b = 1/8 + 1/8 = 2/8 \quad R_b = 4$$

$$V_b = 8/(8+8) \times 3 = 1.5V$$

$$R3' = 4 + 4 = 8$$

Replace $R3'$ and $R2$ by R_c

$$1/R_c = 1/8 + 1/8 = 1/4, \quad R_c = 4$$

$$V_c = 0.5 \times 1.5 = 0.75 \text{ V}$$

$$\text{Finally } I = (6 - 0.75) / (R_1 + R_c) = 5.25 / 10 = 0.525 \text{ A}$$

5 (short)

(a)

$$240 = I(R + jX), \text{ but power factor of } 0.707 \text{ means } X = R.$$

$$\text{so, } IR = 240 / (\sqrt{2})$$

$$P = I^2 R = 24,000, \text{ so } I = 100\sqrt{2}, \text{ } R = 240/200 = 1.2 \text{ ohm}$$

$$P_{\text{line}} = I^2 \cdot 0.02 = \mathbf{400 \text{ W}}.$$

$$V = (1.22^2 + 1.24^2)^{1/2} \cdot 100 \cdot \sqrt{2} = 246 \text{ V}$$

$$Q_{\text{factory}} = 24,000. \text{ } Q_{\text{line}} = 800. \text{ So } Q_{\text{tot}} = 24,800 \text{ VAR}$$

(b)

putting C across factory

$$Q_{\text{fac}} = V^2/X = 24 \times 10^3 \text{ } V = 240 \text{ V}, X = 1/j\omega C = 2.35 \text{ ohm}$$

$$\text{so } C = Q / [2\pi f V^2] = 1.35 \times 10^{-3} \text{ F}.$$

Section B

6 (a) true for $n=1$
for $n=2$

A_1	A_2	$A_1 + A_2$	$\overline{A_1 \cdot A_2}$
0	0	0	0
0	1	1	1
1	0	1	1
1	1	1	1

So true for $n=2$

Prove by induction that this is true for any n

Assume true for $n=k$

$$A_1 + A_2 + \dots + A_k = \overline{\overline{A_1} \cdot \overline{A_2} \dots \overline{A_k}} \quad (1)$$

Show that the expression is true for $n=k+1$

$$\begin{aligned} A_1 + A_2 + \dots + A_k + A_{k+1} &= \overline{\overline{A_1} \cdot \overline{A_2} \dots \overline{A_k}} + A_{k+1} \quad (\text{using (1)}) \\ &= \overline{\overline{A_1} \cdot \overline{A_2} \dots \overline{A_k} \cdot \overline{A_{k+1}}} \end{aligned}$$

(using expression for $n=2$)

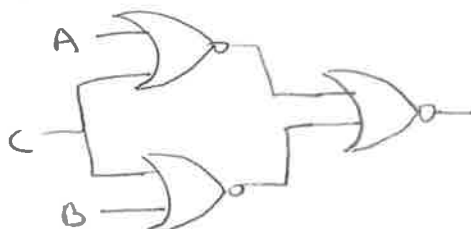
(b) Let $Y = A \cdot B + C$

$$\overline{Y} = \overline{A \cdot B + C}$$

$$= \overline{A+B} \cdot \overline{C}$$

$$\Rightarrow Y = \overline{\overline{A+B} \cdot \overline{C}}$$

C	A B			
	00	01	11	10
0	0	0	1	0
1	1	1	1	1



7

(a) (bookwork)

Static hazard - signal undergoes momentary change when it is supposed to remain unchanged

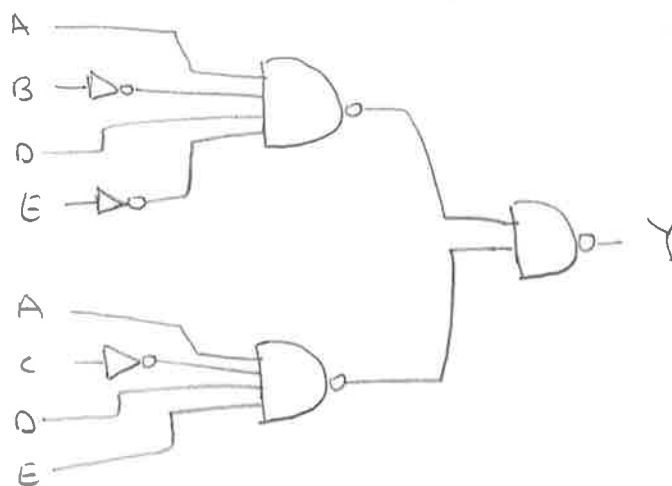
Dynamic hazard - signal changes more than once when it is supposed to change only once

(b)

	AB				$\bar{E}$				E			
CD	00	01	11	10	00	01	11	10	00	01	11	10
00												
01				1							1	1
11				1								
10												

$$Y = A\bar{B}.D.\bar{E} + A\bar{C}.D.E$$

$$= \overline{A\bar{B}.D.\bar{E}} \cdot \overline{A\bar{C}.D.E}$$



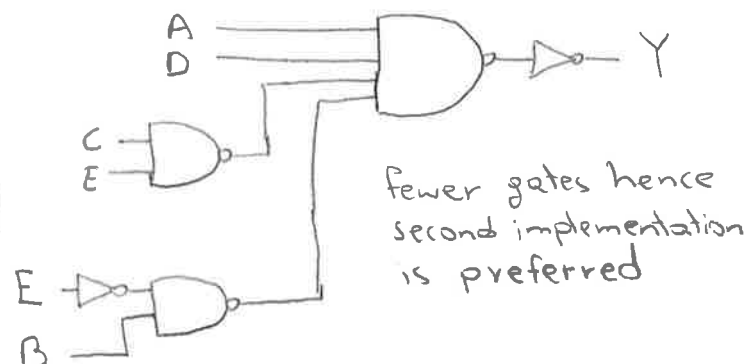
Mapping the inverse of the output

$$\bar{Y} = \bar{A} + \bar{D} + C.E + B.\bar{E}$$

$$= \overline{A.D} + C.E + B.\bar{E}$$

$$= \overline{A.D.C.E.B.\bar{E}}$$

$$\therefore Y = \overline{A.D.C.E.B.\bar{E}}$$



(c) Add the additional term $A\bar{B}.\bar{C}.D$ in the first expression, i.e. $Y = A\bar{B}.D.\bar{E} + A\bar{C}.D.E + A\bar{B}.\bar{C}.D$

8 (a) a : number of address wires
d : number of data wires

$$\begin{aligned} \text{Capacity in bits} &= d \times 2^a \\ &= 8 \times 2^{30} = 2^{33} \end{aligned}$$

1 Mbyte = 10^6 bytes (2^{20} was also perceived as acceptable)

$$\text{So } \underset{\substack{\uparrow \\ \text{1 byte}}}{8 \times 2^{30}} \text{ bits} = \frac{2^{30}}{10^6} \text{ Mbytes} = 1074 \text{ Mbytes}$$

(b) (i)	movlw 0x02	load 2 into register W
	movwf 0x20	load W into memory location 0x20
1b	decfsz 0x20	decrement the contents of mem. location, 0x20. Skip next line if 0
	goto 1b	go to 1b

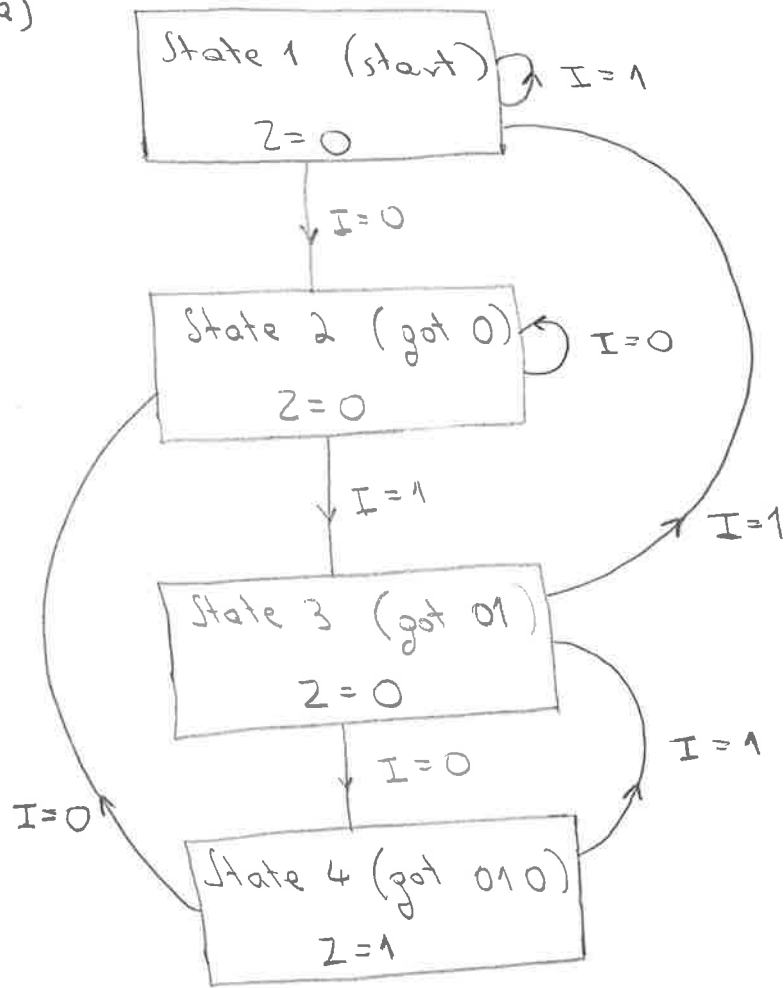
Value 2 is loaded into 0x20 via first 2 lines.
This is then decremented twice in the loop.
After this answer is zero so final line is skipped.

(ii) Value 2^{loaded} into 0x20 via first two lines.
Next line decrements 0x20 (i.e. contents = 1)

goto lb
lb movwf 0x20
will load the contents of W (i.e. 2) into 0x20
Hence the loop will keep repeating itself
without terminating.

9

(a)



(b)

Input I	Current State		Next State		Bistable Inputs			
	Q _A	Q _B	Q _A	Q _B	J _A	K _A	J _B	K _B
0	0	0	0	1	0	x	1	x
1	0	0	0	0	0	x	0	x
0	0	1	0	1	0	x	x	0
1	0	1	1	0	1	x	x	1
0	1	0	1	1	x	0	1	x
1	1	0	0	0	x	1	0	x
0	1	1	0	1	x	1	x	0
1	1	1	1	0	x	0	x	1

(c) J_A $Q_A Q_B$

		00	01	11	10
I	0	0	0	x	x
	1	0	1	x	x

$$J_A = I \cdot Q_B$$

K_A $Q_A Q_B$

		00	01	11	10
I	0	x	x	1	0
	1	x	x	0	1

$$K_A = \bar{I} \cdot Q_B + \bar{Q}_B \cdot I$$

J_B $Q_A Q_B$

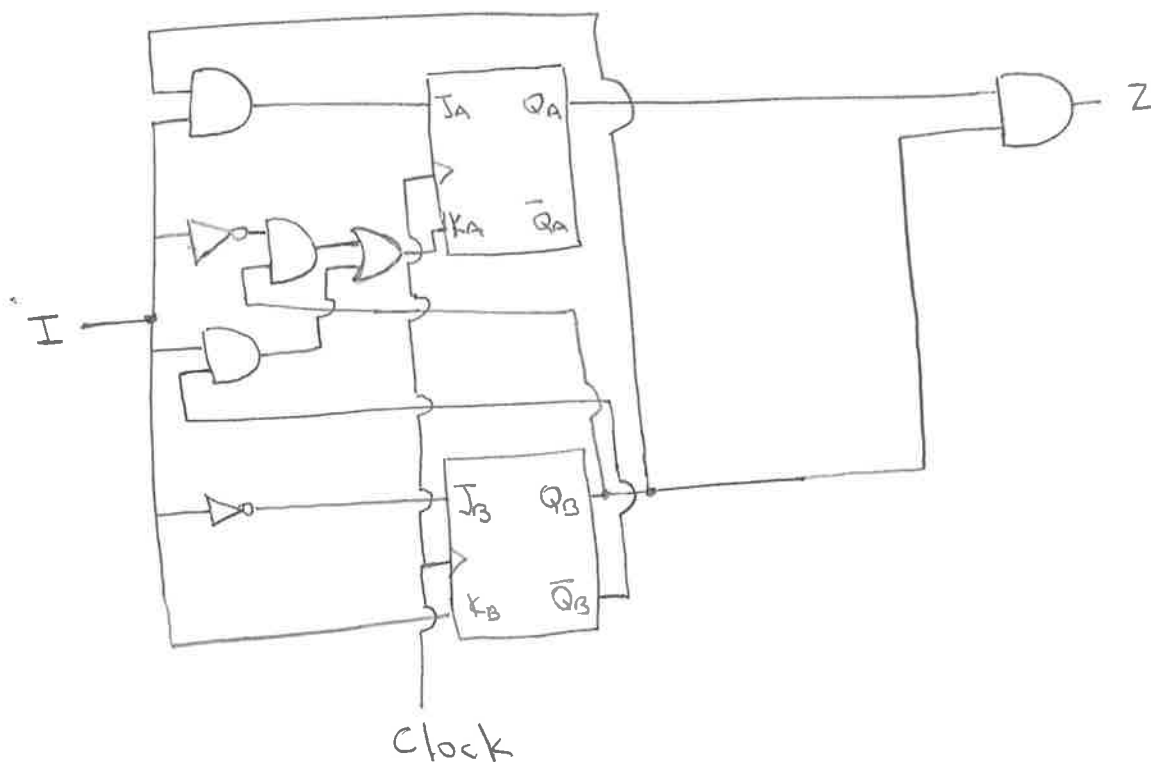
		00	01	11	10
I	0	1	x	x	1
	1	0	x	x	0

$$J_B = \bar{I}$$

K_B $Q_A Q_B$

		00	01	11	10
I	0	x	0	0	x
	1	x	1	1	x

$$K_B = I$$



(d) (i) 2 bistables

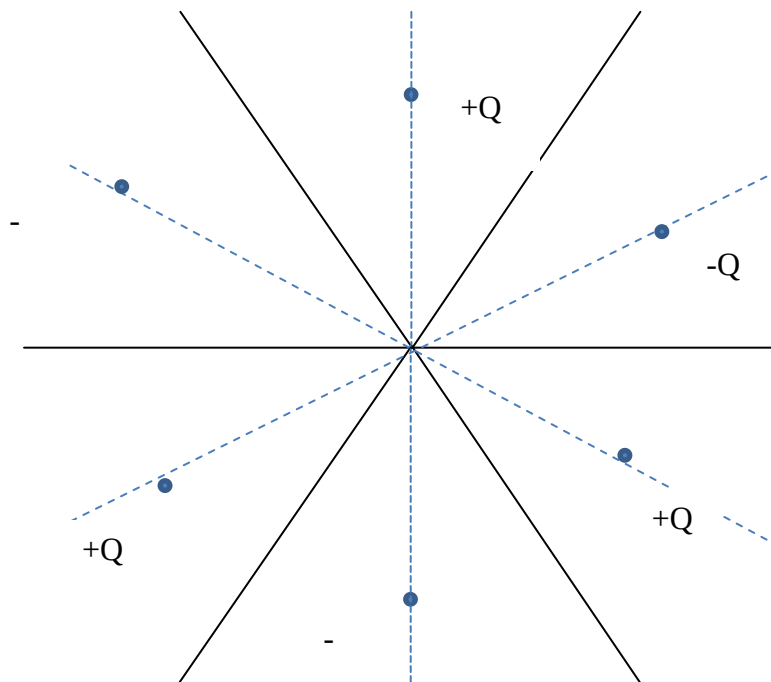
(ii) 2 bistables

SECTION C

10 (short)

(a) Method of images: given that the solution to Poisson's equation is unique, one can use fictitious charges (image charges) to calculate the electric potential and electric forces anywhere if the resulting potential is shown to match all the boundary conditions. For example, a metal sheet is an equipotential, and for a single charge in front of a metal sheet, the image charge of the opposite sign but equal magnitude at the mirror position with respect to the front edge of the sheet produces a zero equipotential at the sheet.

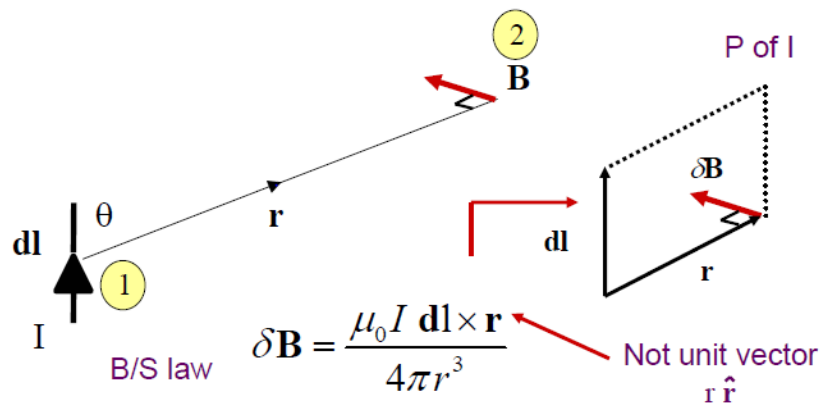
(b) Three sheets at 60 degrees, in pairs, the six charges ensure that the three lines are all equipotentials and so the six charges give the potential profile within the wedge.



(c) The charge will move towards the nearest point where the two sheets meet – i.e. vertically down in Figure 5.

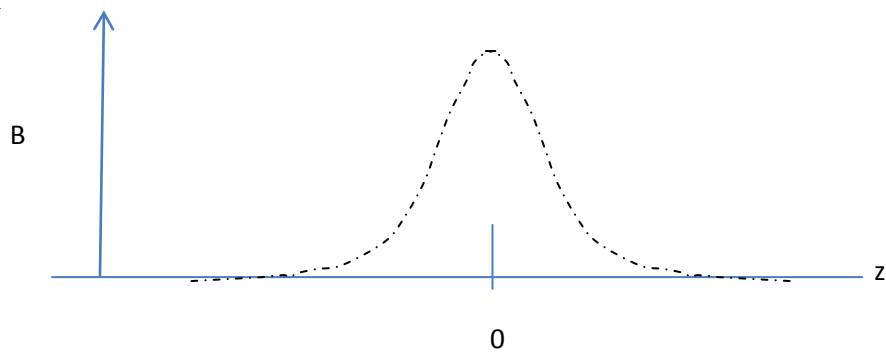
11 (short) Bio-Savart Law:

(a)



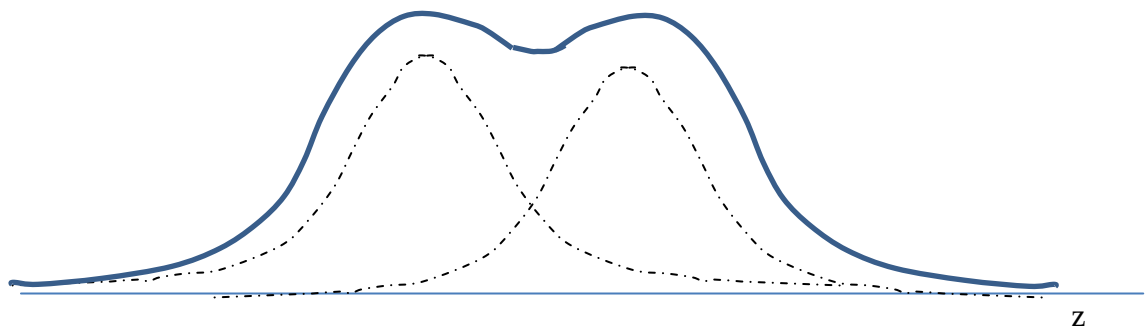
And: $B(z) = \mu_0 I a^2 / 2 [z^2 + a^2]^{3/2}$

Sketch



Note that $B(z)$ is even about z : if you make an error in calculating this result and get $B(z)$ odd in z , the rest of the question makes no sense!

(b)



(c) The field is symmetrical about the central point, so all odd derivatives are zero, and since B'' is zero, this means that the field is very uniform: $B=B_0 + O(z^4)$ near the central point. This is needed when measuring the magnetic properties of materials to ensure that the materials sample sees exactly the same field. It also means that the positioning of the sample is not critical.

12 (long)

(a) Suppose the aluminium bar moves down a distance dx .

If the force is F then the work done is $dW=Fdx$.

A volume of Adx is converted from the field $\mu_0 H$ in the vacuum to $1.000023\mu_0 H$ in aluminium.

The change in energy density is therefore $B^2/2(1.000023\mu_0) - B^2/2\mu_0$
or $-[2.3 \times 10^{-5}]B^2/2\mu_0$.

Thus $Fdx = - [2.3 \times 10^{-5}] AB^2 dx/(2\mu_0)$.

There is an upward force of $[2.3 \times 10^{-5}]10^{-4} (0.5)^2[2 \times 4\pi \times 10^{-7}] = 2.28 \times 10^{-4} \text{ N}$.

[Note many candidates did not take off the energy that was already in the field without the Al being present and got a very large number!]

(b) The current through a resistor during a change of flux because a coil is removed from a magnetic field is given by $I = -[1/R] d\Phi/dt$

If we integrate this over time, the total charge is $Q=\Delta\Phi/R = AB/R$ Coulombs

For typical values: $AB= 10^{-4} 0.5/R = 5 \times 10^{-3}/R = 0.05\mu\text{C}$ for $R=1\text{k}\Omega$

This is a very large charge. Store the change in a capacitor and measure the voltage. Electrolyse material or use a dividing electroscope.

(c) $\text{Emf} = -d\Phi/dt = (d/dt)(NAB \sin\omega t) = AB\omega = 10^{-4} 0.5 2\pi 60 = 0.018\text{V}$ amplitude

(d) The emf is much easier to measure precisely, and the coil can be moved while rotating so the process is much more effective in mapping out the field. For the charge process, one has to be certain that $B=0$, or B is known precisely at the end point of the motion.